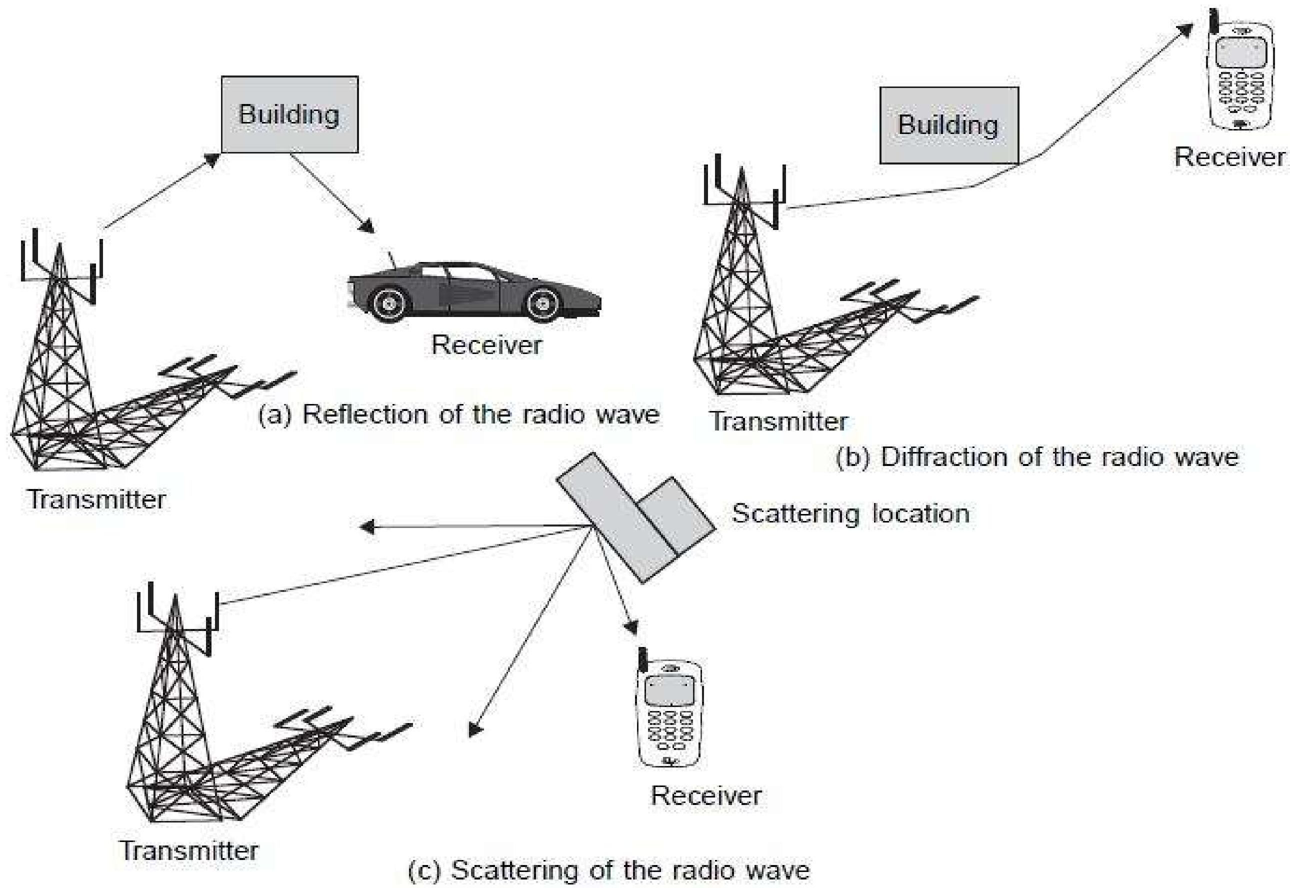
Wireless Propagation

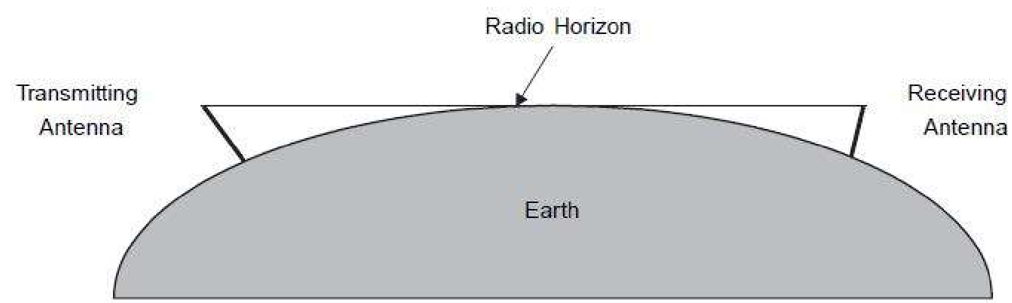
- Radio wave propagation consists of three main attributes: reflection, diffraction and scattering



Wireless Propagation

- Wireless Propagation: large scale and small scale
- Large scale variation of signal occurs due to distance and shadowing
- Small scale variation of signal occurs due to the random movement of environment particles
- Large scale models: free space, path loss model, two ray ground reflection model and path loss with shadowing
- Small scale models: Rayleigh and Rican fading models

Free Space Propagation Model



$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

where P_t is the transmitted power, $P_r(d)$ is the received power which is a function of the T-R separation, G_t is the transmitter antenna gain, G_r is the receiver antenna gain, d is the T-R separation distance in meters, L is the system loss

Free Space Propagation Model

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

- Path loss with antenna gain

$$PL\,(\mathrm{dB}) = 10\log\frac{P_t}{P_r} = -10\log\left[\frac{G_t G_r \lambda^2}{(4\pi)^2 d^2}\right]$$

- Path loss without antenna gain

$$PL\,(\mathrm{dB}) = 10\log\frac{P_t}{P_r} = -10\log\left[\frac{\lambda^2}{(4\pi)^2 d^2}\right]$$

Free Space Propagation Model

- Radiation fields: near filed and far field

- Far-field distance $d_f = \frac{2D^2}{\lambda}$

  where D is the largest physical linear dimension of the antenna

- Conditions of far field

  $d_f \gg D$ $\qquad$ $d_f \gg \lambda$

- Free space model can be rewritten as

$$P_r(d) = P_r(d_0)\left(\frac{d_0}{d}\right)^2 \qquad d \geq d_0 \geq d_f$$

Example 3.2

If a transmitter produces 50 watts of power, express the transmit power in units of (a) dBm, and (b) dBW. If 50 watts is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is P_r (10 km) ? Assume unity gain for the receiver antenna.

Solution to Example 3.2

Given:

Transmitter power, $P_t = 50$ W.

Carrier frequency, $f_c = 900$ MHz

Using equation (3.9),

(a) Transmitter power,

$$P_t(\text{dBm}) = 10\log[P_t(\text{mW})/(1\text{ mW})]$$
$$= 10\log[50 \times 10^3] = 47.0 \text{ dBm.}$$

(b) Transmitter power,

$$P_t(\text{dBW}) = 10\log[P_t(\text{W})/(1\text{ W})]$$
$$= 10\log[50] = 17.0 \text{ dBW.}$$

The received power can be determined using equation (3.1).

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L} = \frac{50(1)(1)(1/3)^2}{(4\pi)^2(100)^2(1)} = 3.5 \times 10^{-6} \text{ W} = 3.5 \times 10^{-3} \text{ mW}$$

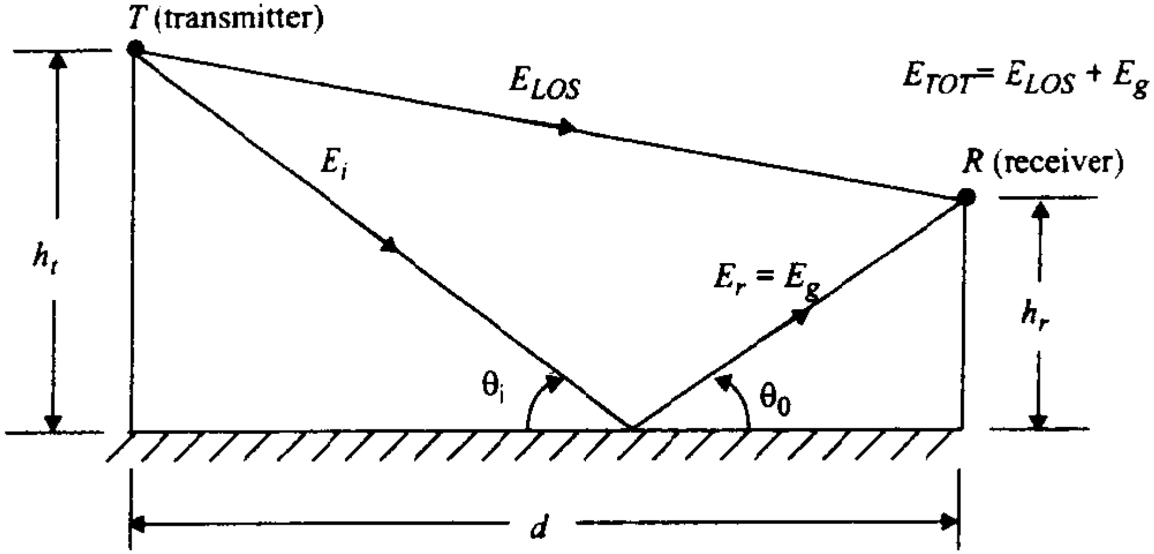
$$P_r(\text{dBm}) = 10\log P_r(\text{mW}) = 10\log\left(3.5 \times 10^{-3} \text{ mW}\right) = -24.5 \text{ dBm.}$$

The received power at 10 km can be expressed in terms of dBm using equation (3.9), where $d_0 = 100$ m and $d = 10$ km

$$P_r(10\text{ km}) = P_r(100) + 20\log\left[\frac{100}{10000}\right] = -24.5 \text{ dBm} - 40 \text{ dB}$$

$$= -64.5 \text{ dBm.}$$

Two-Ray Ground Reflection Model



$$\theta_i = \theta_0$$

If E_0 is the free space E-field (in units of V/m) at a reference distance d_0 from the transmitter, then for $d > d_0$

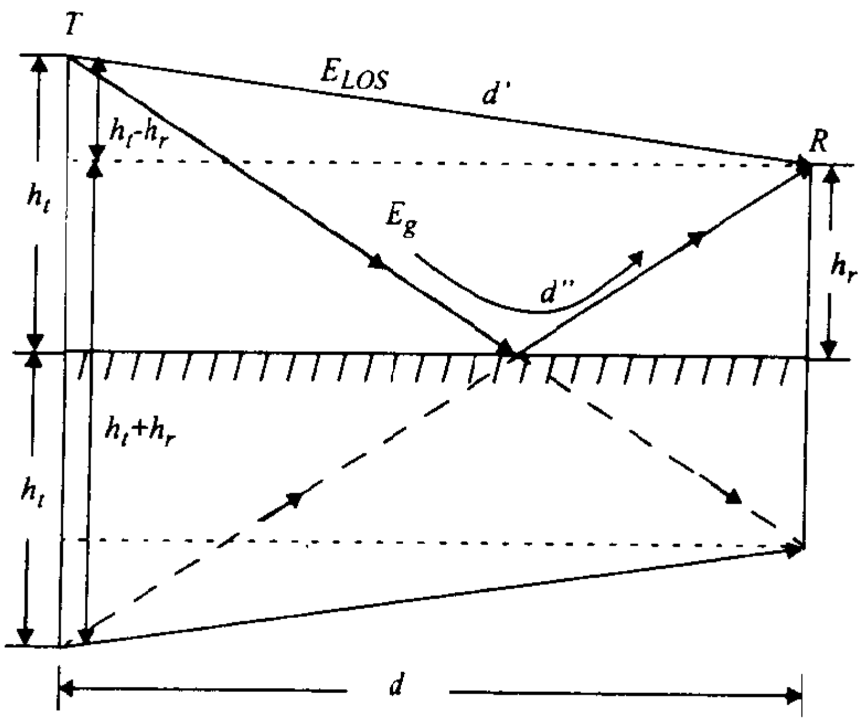
$$E(d,t) = \frac{E_0 d_0}{d}\cos\left(\omega_c\left(t - \frac{d}{c}\right)\right) \qquad (d > d_0)$$

$$E_{LOS}(d',t) = \frac{E_0 d_0}{d'}\cos\left(\omega_c\left(t - \frac{d'}{c}\right)\right)$$

$$E_g(d'',t) = \Gamma\frac{E_0 d_0}{d''}\cos\left(\omega_c\left(t - \frac{d''}{c}\right)\right)$$

Two-Ray Ground Reflection Model

$$E_{TOT}(d,t) = \frac{E_0 d_0}{d'}\cos\left(\omega_c\left(t-\frac{d'}{c}\right)\right) + (-1)\frac{E_0 d_0}{d''}\cos\left(\omega_c\left(t-\frac{d''}{c}\right)\right)$$



$$\Delta = d'' - d' \approx \frac{2h_t h_r}{d}$$

$$\Delta = d'' - d' = \sqrt{(h_t+h_r)^2 + d^2} - \sqrt{(h_t-h_r)^2 + d^2}$$

Two-Ray Ground Reflection Model

$$\theta_\Delta = \frac{2\pi\Delta}{\lambda} = \frac{\Delta\omega_c}{c} \qquad\qquad \tau_d = \frac{\Delta}{c} = \frac{\theta_\Delta}{2\pi f_c}$$

$$E_{TOT}(d,t) = \frac{E_0 d_0}{d'}\cos\left(\omega_c\left(t-\frac{d'}{c}\right)\right) + (-1)\frac{E_0 d_0}{d''}\cos\left(\omega_c\left(t-\frac{d''}{c}\right)\right)$$

- Assuming,

$$\left|\frac{E_0 d_0}{d}\right| \approx \left|\frac{E_0 d_0}{d'}\right| \approx \left|\frac{E_0 d_0}{d''}\right|$$

$$\begin{aligned} E_{TOT}\left(d, t = \frac{d''}{c}\right) &= \frac{E_0 d_0}{d'}\cos\left(\omega_c\left(\frac{d''-d'}{c}\right)\right) - \frac{E_0 d_0}{d''}\cos 0^o \\ &= \frac{E_0 d_0}{d'}\cos\theta_\Delta - \frac{E_0 d_0}{d''} \\ &\approx \frac{E_o d_0}{d}\left[\cos\theta_\Delta - 1\right] \end{aligned}$$

Two-Ray Ground Reflection Model

$$|E_{TOT}(d)| = \sqrt{\left(\frac{E_0 d_0}{d}\right)^2 (\cos\theta_\Delta - 1)^2 + \left(\frac{E_0 d_0}{d}\right)^2 \sin^2\theta_\Delta}$$

$$|E_{TOT}(d)| = \frac{E_0 d_0}{d}\sqrt{2 - 2\cos\theta_\Delta}$$

$$|E_{TOT}(d)| = 2\frac{E_0 d_0}{d}\sin\left(\frac{\theta_\Delta}{2}\right)$$

$$\theta_\Delta = \frac{2\pi\Delta}{\lambda} = \frac{\Delta\omega_c}{c}$$

$$E_{TOT}(d) \approx \frac{2E_0 d_0}{d}\frac{2\pi h_t h_r}{\lambda d}$$

$$\Delta = d'' - d' \approx \frac{2h_t h_r}{d}$$

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}$$

$$E_0^2 = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d_0^{\ 2}}$$

Example 3.1

With $h_b = 100\,\text{ft}$, and $h_m = 5\,\text{ft}$, and a frequency of 881.52 MHz ($\lambda = 1.116\,\text{ft}$), calculate signal attenuation at a distance equal to 5000 ft. Assume antenna gains are 8 dB and 0 dB for the base station and mobile station, respectively. What are the free-space and reflected surface attenuations? Assume the earth surface to be flat.

Solution

$$\bar{d} = \frac{4h_b h_m}{\lambda} = \frac{4 \times 100 \times 5}{1.116} = 1792\,\text{ft},\ d > \bar{d}$$

$$G_b = 8\,\text{dB} = 10^{0.8} = 6.3; \quad G_m = 0\,\text{dB} = 1.0$$

Free-space attenuation:

$$\frac{P_t}{P_r} = \left(\frac{4\pi d}{\lambda}\right)^2 \cdot \frac{1}{G_b G_m} = \left(\frac{4\pi \times 5000}{1.116}\right)^2 \cdot \frac{1}{6.3 \times 1} = 87\,\text{dB}$$

Attenuation on reflecting surface:

$$\frac{P_t}{P_r} = \left[\frac{d^2}{h_b h_m}\right]^2 \cdot \frac{1}{G_b G_m} = \left[\frac{5000^2}{100 \times 5}\right]^2 \cdot \frac{1}{6.3 \times 1} = 86\,\text{dB}$$

Log Distance Path Loss Model

- In natural unit:

$$P_r(d) = P_r(d_0)\left(\frac{d_0}{d}\right)^n \qquad d \ge d_0 \ge d_f$$

$$P_r(d_0) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d_0^{\,2}}$$

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d_0^{\,2}}\left(\frac{d_0}{d}\right)^\eta$$

- In dBm unit:

$$P_r(d)[dBm] = 10\log[1000 * P_r(d)]$$

$$= 10\log\left[1000 * \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d_0^{\,2}}\left(\frac{d_0}{d}\right)^\eta\right]$$

$$= 10\log(1000 * P_t) + 10\log\left[\frac{G_t G_r \lambda^2}{(4\pi)^2 d_0^{\,2}}\right] + 10\log\left(\frac{d_0}{d}\right)^\eta$$

$$= P_t[dBm] - \overline{PL(d_0)} - 10\log\left(\frac{d}{d_0}\right)^\eta$$

Path loss at reference or received power at reference distance is calculated by free space model

$$\overline{PL(d_0)} = -10\log_{10}\left(\frac{G_t G_r \lambda^2}{(4\pi)^2 d_0^{\,2}}\right)$$

Log Distance Path Loss Model

$$P_r(d)[dBm] = P_t[dBm] - \overline{PL(d_0)} - 10\log\left(\frac{d}{d_0}\right)^{\eta}$$

$$= P_t[dBm] - \left[\overline{PL(d_0)} + 10\log\left(\frac{d}{d_0}\right)^{\eta}\right]$$

$$\overline{PL}(\mathrm{dB}) = \overline{PL}(d_0) + 10n\log\left(\frac{d}{d_0}\right)$$

$$P_r(d)[dBm] = P_t[dBm] - PL(d)[dB]$$

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

Problem

- Determine received power at distance 1 km under log distance path loss model with path loss exponents 3. Assume, transmit power = 10 W, transmitter antenna gain = 6 dB, receiver antenna gain= 4 dB, frequency = 1800 MHz, and reference distance = 10 m

Solution:

Gt=10^(6/10)=3.98

Gr=10^(4/10)=2.51

Lambda=1/6

Pt=10*log(10000)=40 dBm

PL=PL(d0)+10*3*log(1000/10) =-10*log([3.98*2.51]/[36*100*16*pi^2)+60=44.14+60 =104.14

Pr=40-104.14=-64.14 dBm

Log Distance Path Loss Model with Log Normal Shadowing

$$PL(d)[dB] = \overline{PL}(d) + X_\sigma = \overline{PL}(d_0) + 10n\log\left(\frac{d}{d_0}\right) + X_\sigma$$

where X_σ is a zero-mean Gaussian distributed random variable (in dB) with standard deviation σ (also in dB).

$$P_r(d)[dBm] = P_t[dBm] - PL(d)[dB]$$

Gaussian probability distribution function (PDF): $p(y) = \Pr(X_\sigma = y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$

$$\Pr(a < X_\sigma < b) = \int_a^b p(y)dy = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy$$

Outage Probability:

$$\begin{aligned}
&\Pr\left(P_r(d) < \gamma\right) \\
&= \Pr\left(P_t - PL(d) < \gamma\right) \\
&= \Pr\left(P_t - \overline{PL}(d_0) - 10\eta\log(\frac{d}{d_0}) - X_\sigma < \gamma\right)
\end{aligned}$$

$$\overline{P_r(d)} = P_t - \overline{PL(d_0)} - 10\eta\log(\frac{d}{d_0})$$

$$\Pr\left(P_r(d) < \gamma\right) = \Pr\left(\overline{P_r}(d) - X_\sigma < \gamma\right) = \Pr\left(X_\sigma > \overline{P_r}(d) - \gamma\right)$$

Outage Probability

$$\Pr\left(P_r(d) < \gamma\right) = \Pr\left(X_\sigma > \overline{P_r}(d) - \gamma\right)$$

$$= \int_{\overline{P_r}(d)-\gamma}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{y^2}{\sigma^2}\right) dy$$

$$= \int_{\frac{\overline{P_r}(d)-\gamma}{\sqrt{2}\sigma}}^{\infty} \frac{1}{\sqrt{\pi}} \exp(-z^2) dz$$

$$= \frac{1}{2} erfc\left(\frac{\overline{P_r}(d) - \gamma}{\sqrt{2}\sigma}\right)$$

$$Pr[P_r(d) < \gamma] = Q\left(\frac{\overline{P_r(d)} - \gamma}{\sigma}\right)$$

$$p(y) = \Pr(X_\sigma = y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

$$\mathrm{erf}(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-y^2} dy$$

$$\mathrm{erfc}(u) = \frac{2}{\sqrt{\pi}} \int_u^\infty e^{-y^2} dy$$

$$Q(u) = \frac{1}{2\sqrt{\pi}} \int_u^\infty e^{-y^2} dy$$

$$Q(u) = \frac{1}{2} \mathrm{erfc}\left[\frac{u}{\sqrt{2}}\right]$$

TABLE C.1 Values of $Q(x)$ versus x.

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.00990	.00964	.00939	.00914	.00889	.00866	.00842
2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139

TABLE C.2 Values of $Q(x)$ for large values of x.

x	10 log x	$Q(x)$	x	10 log x	$Q(x)$	x	10 log x	$Q(x)$
3.00	4.77	1.35E-03	4.00	6.02	3.17E-05	5.00	6.99	2.87E-07
3.05	4.84	1.14E-03	4.05	6.07	2.56E-05	5.05	7.03	2.21E-07
3.10	4.91	9.68E-04	4.10	6.13	2.07E-05	5.10	7.08	1.70E-07
3.15	4.98	8.16E-04	4.15	6.18	1.66E-05	5.15	7.12	1.30E-07
3.20	5.05	6.87E-04	4.20	6.23	1.33E-05	5.20	7.16	9.96E-08
3.25	5.12	5.77E-04	4.25	6.28	1.07E-05	5.25	7.20	7.61E-08
3.30	5.19	4.83E-04	4.30	6.33	8.54E-06	5.30	7.24	5.79E-08
3.35	5.25	4.04E-04	4.35	6.38	6.81E-06	5.35	7.28	4.40E-08
3.40	5.31	3.37E-04	4.40	6.43	5.41E-06	5.40	7.32	3.33E-08
3.45	5.38	2.80E-04	4.45	6.48	4.29E-06	5.45	7.36	2.52E-08
3.50	5.44	2.33E-04	4.50	6.53	3.40E-06	5.50	7.40	1.90E-08
3.55	5.50	1.93E-04	4.55	6.58	2.68E-06	5.55	7.44	1.43E-08
3.60	5.56	1.59E-04	4.60	6.63	2.11E-06	5.60	7.48	1.07E-08
3.65	5.62	1.31E-04	4.65	6.67	1.66E-06	5.65	7.52	8.03E-09
3.70	5.68	1.08E-04	4.70	6.72	1.30E-06	5.70	7.56	6.00E-09
3.75	5.74	8.84E-05	4.75	6.77	1.02E-06	5.75	7.60	4.47E-09
3.80	5.80	7.23E-05	4.80	6.81	7.93E-07	5.80	7.63	3.32E-09
3.85	5.85	5.91E-05	4.85	6.86	6.17E-07	5.85	7.67	2.46E-09
3.90	5.91	4.81E-05	4.90	6.90	4.79E-07	5.90	7.71	1.82E-09
3.95	5.97	3.91E-05	4.95	6.95	3.71E-07	5.95	7.75	1.34E-09

Example 3.3

Calculate the received power at a distance of 3 km from the transmitter if the path-loss exponent γ is 4. Assume the transmitting power of 4 W at 1800 MHz, a shadow effect of 10.5 dB, and the power at reference distance ($d_0 = 100\,\text{m}$) of -32 dBm. What is the allowable path loss?

Solution

Using Equation 3.11 and including shadow effect we get

$$P_r(d) = 10\log[P_0(d_0)] + 10\gamma\log(d_0/d) + X_\sigma$$

$$\therefore P_r = -32 + 10 \times 4 \times \log\left(\frac{100}{3000}\right) + 10.5 = -80.5\,\text{dBm}$$

$$\text{Allowable path loss} = P_t - P_r = 36 - (-80.5) = 116.5\,\text{dB}$$

Example 3.4

What is the separation distance between the transmitter and the receiver with an allowable path loss of 150 dB and shadow effect of 10 dB? The path loss in dB is given as:

$$L_p = 133.2 + 43 \log d$$

where:

d = separation distance in km

Solution

Using Equation 3.13, we have

$$150 = 133.2 + 40 \log d + 10$$

$$\log d = \frac{6.8}{40} = 0.17$$

$$d = 10^{0.17} = 1.48 \text{ km}$$

Small Scale Multi-path Propagation

- Multipath propagation creates small scale fading effect
- Three main effects:
  - Rapid changes in signal strength over a small travel distance or time interval
  - Random frequency modulation due to varying Doppler shifts on different multipath signals
  - Time dispersion (echoes) caused by multipath propagation delays.

Factors Influence Small Scale Fading

- Multipath Propagation
  - Signals reach to destination with different phase, time and amplitude
  - Dispersion and inter symbol interference (ISI) occur
- Speed of mobile
  - Relative speed results in random FM due to doppler shifts
- Speed of surrounding objects
  - Relative speed of multipath objects results in random FM due to doppler shifts
- Transmission bandwidth of the signal
- If bandwidth of channel is lower than signal bandwidth; the signal distorted

Doppler Shift

- The difference of propagation path length

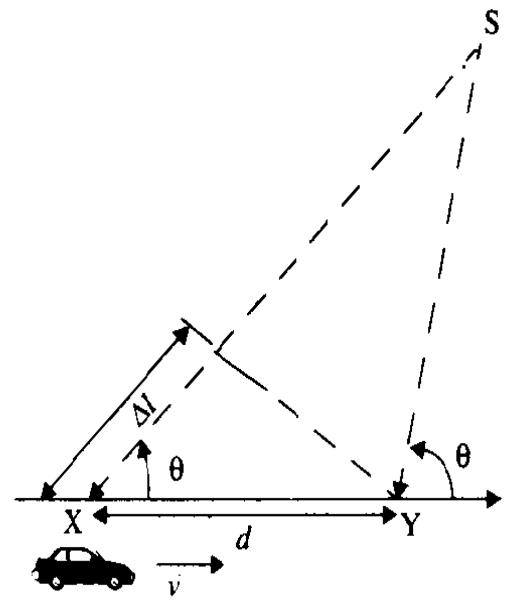
$$\Delta l = d\cos\theta = v\Delta t\cos\theta$$

- Phase change in received signal

$$\Delta\phi = \frac{2\pi\Delta l}{\lambda} = \frac{2\pi v\Delta t}{\lambda}\cos\theta$$

- Change in frequency

$$f_d = \frac{1}{2\pi}\cdot\frac{\Delta\phi}{\Delta t} = \frac{v}{\lambda}\cdot\cos\theta$$



- Thus, the carrier frequency continuously changes and results in FM

Example 4.1

Consider a transmitter which radiates a sinusoidal carrier frequency of 1850 MHz. For a vehicle moving 60 mph, compute the received carrier frequency if the mobile is moving (a) directly towards the transmitter, (b) directly away from the transmitter, (c) in a direction which is perpendicular to the direction of arrival of the transmitted signal.

Solution to Example 4.1

Given:

Carrier frequency $f_c = 1850\,\mathrm{MHz}$

Therefore, wavelength $\lambda = c/f_c = \dfrac{3 \times 10^8}{1850 \times 10^6} = 0.162\,\mathrm{m}$

Vehicle speed $v = 60$ mph $= 26.82$ m/s

(a) The vehicle is moving directly towards the transmitter.
The Doppler shift in this case is positive and the received frequency is given by equation (4.2)

$$f = f_c + f_d = 1850 \times 10^6 + \frac{26.82}{0.162} = 1850.00016\ \mathrm{MHz}$$

(b) The vehicle is moving directly away from the transmitter.
The Doppler shift in this case is negative and hence the received frequency is given by

$$f = f_c - f_d = 1850 \times 10^6 - \frac{26.82}{0.162} = 1849.999834 \text{ MHz}$$

(c) The vehicle is moving perpendicular to the angle of arrival of the transmitted signal.

In this case, $\theta = 90°$, $\cos\theta = 0$, and there is no Doppler shift.

The received signal frequency is the same as the transmitted frequency of 1850 MHz.

Autocorrelation and Convolution

The correlation of a signal with itself is its *autocorrelation*:

$$w(t) = u(t) \otimes u(t) = \int_{-\infty}^{\infty} u^*(\tau - t) u(\tau) d\tau$$

The autocorrelation at zero lag:

$$\begin{aligned} w(0) &= \int_{-\infty}^{\infty} u^*(\tau - 0) u(\tau) d\tau \\ &= \int_{-\infty}^{\infty} u^*(\tau) u(\tau) d\tau \\ &= \int_{-\infty}^{\infty} |u(\tau)|^2 d\tau = E_u \end{aligned}$$

The autocorrelation at zero lag, $w(0)$, is the energy of the signal.

The continuous-time convolution of two signals $f_1(t)$ and $f_2(t)$

$$f(t) = f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

- **If the impulse response of a system is h(t); then output y(t) for a given input x(t): y(t)=h(t)*x(t)**

Probability, Mean and Variance

- Let X is a random variable
- Probability distribution function (PDF) of X , i.e., Pr(X=x) is a function of x that represents the probability that the value of X be x
- Mean/average value of X:

$$\mu = E[X] = \int_0^{\infty} x \Pr(X = x) dx$$

- Variance/ mean square of X:

$$\sigma^2 = \int_0^{\infty} (x - \mu)^2 \Pr(X = x) dx$$

Time-Varying Channel Impulse Response

- Let the transmitted signal

$$s(t) = \Re\left\{u(t)e^{j(2\pi f_c t+\phi_0)}\right\} = \Re\{u(t)\}\cos(2\pi f_c t+\phi_0) - \Im\{u(t)\}\sin(2\pi f_c t+\phi_0),$$

- The corresponding received signal is the sum of the line-of-sight (LOS) path and all resolvable multipath components

$$r(t) = \Re\left\{\sum_{n=0}^{N(t)} \alpha_n(t)u(t-\tau_n(t))e^{j2\pi[f_c(t-\tau_n(t))+\phi_{D_n}+\phi_0]}\right\}$$

$$\phi_n(t) = 2\pi f_c \tau_n(t) - \phi_{D_n} - \phi_0.$$

$$r(t) = \Re\left\{\left[\sum_{n=0}^{N(t)} \alpha_n(t)e^{-j\phi_n(t)}u(t-\tau_n(t))\right]e^{j2\pi f_c t}\right\}.$$

Time-Varying Channel Impulse Response

$$r(t) = \Re\left\{ \left(\int_{-\infty}^{\infty} c(\tau, t) u(t-\tau) d\tau \right) e^{j2\pi f_c t} \right\}$$

$$c(\tau, t) = \sum_{n=0}^{N(t)} \alpha_n(t) e^{-j\phi_n(t)} \delta(\tau - \tau_n(t)),$$

$$r(t) = \Re\left\{ \left[\sum_{n=0}^{N(t)} \alpha_n(t) e^{-j\phi_n(t)} u(t - \tau_n(t)) \right] e^{j2\pi f_c t} \right\}.$$

$c(\tau, t)$ has two time parameters: the time t when the impulse response is observed at the receiver,

time $t - \tau$ when the impulse is launched into the channel relative to the observation time t.

Channel autocorrelation functions

- The statistical characterization of channel *is determined by its autocorrelation function*

$$A_c(\tau_1, \tau_2; t, \Delta t) = E[c^*(\tau_1; t) c(\tau_2; t + \Delta t)]$$

- **Wide sense stationary (WSS) channel**

joint statistics of a channel measured at two different times

$t \text{ and } t+\Delta t$ depends only on the time difference Δt

- For WSS channel autocorrelation function

$$A_c(\tau_1, \tau_2; \Delta t) = \quad E[c^*(\tau_1; t) c(\tau_2; t + \Delta t)]$$

WSSUS

- In practice, the channel response associated with a given multipath component of delay τ_1 is uncorrelated with the response associated with a multipath component at a different delay τ_2 since the two components are caused by different scatterers
- Thus, wireless channel has uncorrelated scattering (US) and channels that are WSS with US is known as WSSUS channels
- For a WSSUS channel, autocorrelation function

$$E[c^*(\tau_1;t)c(\tau_2;t+\Delta t)] = A_c(\tau_1;\Delta t)\delta[\tau_1-\tau_2] \stackrel{\triangle}{=} A_c(\tau;\Delta t),$$

Scattering Function

- The scattering function for random channels is defined as the Fourier transform of $A_c(\tau;\Delta t)$ with respect to the Δt parameter:

$$S_c(\tau,\rho) = \int_{-\infty}^{\infty} A_c(\tau,\Delta t) e^{-j2\pi\rho\Delta t} d\Delta t.$$

- The scattering function characterizes the average output power associated with the channel as a function of the multipath delay τ *and Doppler shift* ρ

Power delay profile

- The power delay profile/ the multipath intensity profile is defined as the autocorrelation $\Delta t = 0$: $A_c(\tau) \triangleq A_c(\tau, 0)$
- The power delay profile represents the average power associated with a given multipath delay, and is easily measured empirically

Note that if we define the pdf p_{T_m} of the random delay spread T_m in terms of $A_c(\tau)$ as

$$p_{T_m}(\tau) = p(T_m = \tau) = \frac{A_c(\tau)}{\int_0^\infty A_c(\tau) d\tau}$$

- The average delay $\mu_{T_m} = \frac{\int_0^\infty \tau A_c(\tau) d\tau}{\int_0^\infty A_c(\tau) d\tau}$

- RMS delay $\sigma_{T_m} = \sqrt{\frac{\int_0^\infty (\tau - \mu_{T_m})^2 A_c(\tau) d\tau}{\int_0^\infty A_c(\tau) d\tau}}.$

Example

The power delay spectrum is often modeled as having a one-sided exponential distribution:

$$A_c(\tau) = \frac{1}{\overline{T}_m} e^{-\tau/\overline{T}_m}, \ \tau \geq 0.$$

Show that the average delay spread (3.54) is $\mu_{T_m} = \overline{T}_m$ and find the rms delay spread (3.55).

Solution: It is easily shown that $A_c(\tau)$ integrates to one. The average delay spread is thus given by

$$\mu_{T_m} = \frac{1}{\overline{T}_m} \int_0^\infty \tau e^{-\tau/\overline{T}_m} = \overline{T}_m.$$

$$\sigma_{T_m} = \sqrt{\frac{1}{\overline{T}_m} \int_0^\infty \tau^2 e^{-\tau/\overline{T}_m} d\tau - \mu_{T_m}^2} = 2\overline{T}_m. - \overline{T}_m = \overline{T}_m.$$

$$A_c(\tau) = \frac{1}{\bar{T}_m} e^{-\tau/\bar{T}_m} \quad (\tau \geq 0)$$

$$\mu = \frac{\int_0^\infty \tau \cdot \frac{1}{\bar{T}_m} e^{-\tau/\bar{T}_m}\, d\tau}{\int_0^\infty \frac{1}{\bar{T}_m} e^{-\tau/\bar{T}_m}\, d\tau}$$

$$= \frac{\frac{\bar{T}_m^2}{\bar{T}_m}\left[e^{-\tau/\bar{T}_m}(-\tau-1)\right]_0^\infty}{\left[-e^{-\tau/\bar{T}_m}\right]_0^\infty}$$

$$= \frac{\bar{T}_m \times 1}{1}$$

$$= \bar{T}_m .$$

$$\sigma^2 = \frac{\int_0^\infty (\tau-\mu)^2 A_c(\tau)\, d\tau}{\int_0^\infty A_c(\tau)\, d\tau} = \frac{\int_0^\infty \tau^2 A_c(\tau)\, d\tau - \int 2\mu\tau A_c(\tau)\, d\tau + \mu^2 \int_0^\infty A_c(\tau)\, d\tau}{1}$$

$$= \frac{\int_0^\infty \tau^2 \frac{1}{\bar{T}_m} e^{-\tau/\bar{T}_m}\, d\tau - 2\bar{T}_m^2 + \bar{T}_m^2}{1}$$

$$= \frac{1}{\bar{T}_m}\left[e^{-\tau/\bar{T}_m}\left(\frac{\tau^2}{-\frac{1}{\bar{T}_m}} - \frac{2\tau}{\frac{1}{\bar{T}_m^2}} + \frac{2}{-\frac{1}{\bar{T}_m^3}} \right)\right]_0^\infty - \bar{T}_m^2$$

$$= \frac{2}{\bar{T}_m} \bar{T}_m^3 - \bar{T}_m^2 = \bar{T}_m^2$$

$$\sigma = \bar{T}_m$$

Example

Consider a wideband channel with multipath intensity profile

$$A_c(\tau) = \begin{cases} e^{-\tau} & 0 \leq \tau \leq 20 \ \mu\text{sec.} \\ 0 & \text{else} \end{cases} .$$

Find the mean and rms delay spreads of the channel and find the maximum symbol period such that a linearly-modulated signal transmitted through this channel does not experience ISI.

Solution: The average delay spread is

$$\mu_{T_m} = \frac{\int_0^{20*10^{-6}} \tau e^{-\tau} d\tau}{\int_0^{20*10^{-6}} e^{-\tau} d\tau} = 10 \ \mu\text{sec.}$$

The rms delay spread is

$$\sigma_{T_m} = \frac{\int_0^{20*10^{-6}} (\tau - \mu_{T_m})^2 e^{-\tau} d\tau}{\int_0^{20*10^{-6}} e^{-\tau} d\tau} = 5.76 \ \mu\text{sec}.$$

We see in this example that the mean delay spread is roughly twice its rms value, in contrast to the prior example where they were equal. This is due to the fact that in this example all multipath components have delay spread less than 20 μsec, which limits the rms spread. When the average and rms delay spreads differ significantly we use rms delay spread to determine ISI effects. Specifically, to avoid ISI we require linear modulation to have a symbol period T_s that is large relative to σ_{T_m}. Taking this to mean that $T_s > 10\sigma_{T_m}$ yields a symbol period of $T_s = 57.6$ μsec or a symbol rate of $R_s = 1/T_s = 17.35$ Kilosymbols per second. This is a highly constrained symbol rate for many wireless systems. Specifically, for binary modulations where the symbol rate equals the data rate (bits per second, or bps), high-quality voice requires on the order of 32 Kbps and high-speed data can requires on the order of 10-100 Mbps.

Power-delay Profile for discrete case

- Mean access delay:

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k)\,\tau_k}{\sum_k P(\tau_k)}$$

- RMS delay spread:

$$\sigma_\tau = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}$$

$$\overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k)\,\tau_k^2}{\sum_k P(\tau_k)}$$

Coherence bandwidth and Coherence Time

- Coherence bandwidth is a statistical measure of frequncy range over which the channel can be considered as “Flat”

$$B_c \approx \frac{1}{5\sigma_\tau}$$

- Coherence time is the statistical measure of time duration over which the channel impulse response is time invariant
- Coherence time is related to the Doppler shifts
- If f_m is the maximum doppler shift, coherence time

$$T_C \approx \frac{1}{f_m}$$

- Doppler shift

$$f_d = \frac{1}{2\pi} \cdot \frac{\Delta\phi}{\Delta t} = \frac{v}{\lambda} \cdot \cos\theta$$

Example 4.4

Calculate the mean excess delay, rms delay spread, and the maximum excess delay (10 dB) for the multipath profile given in the figure below. Estimate the 50% coherence bandwidth of the channel. Would this channel be suitable for AMPS or GSM service without the use of an equalizer?

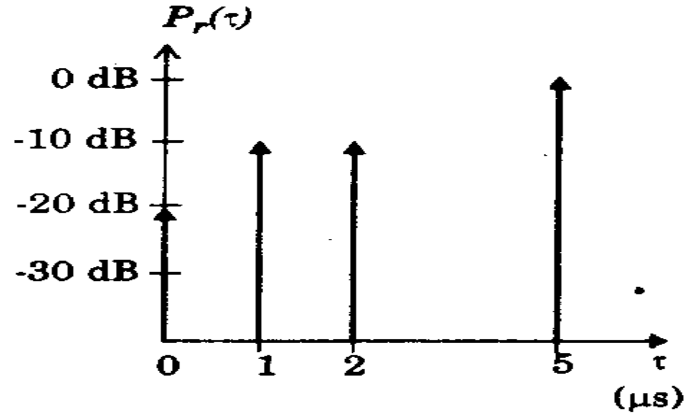


Figure E4.4

Solution to Example 4.4

The rms delay spread for the given multipath profile can be obtained using equations (4.35) — (4.37). The delays of each profile are measured relative to the first detectable signal. The mean excess delay for the given profile

$$\bar{\tau} = \frac{(1)(5) + (0.1)(1) + (0.1)(2) + (0.01)(0)}{[0.01 + 0.1 + 0.1 + 1]} = 4.38\mu s$$

The second moment for the given power delay profile can be calculated as

$$\overline{\tau^2} = \frac{(1)(5)^2 + (0.1)(1)^2 + (0.1)(2)^2 + (0.01)(0)}{1.21} = 21.07\mu s^2$$

Therefore the rms delay spread, $\sigma_\tau = \sqrt{21.07 - (4.38)^2} = 1.37\mu s$

The coherence bandwidth is found from equation (4.39) to be

$$B_c \approx \frac{1}{5\sigma_\tau} = \frac{1}{5(1.37\mu s)} = 146 \text{ kHz}$$

Since B_c is greater than 30 kHz, AMPS will work without an equalizer. However, GSM requires 200 kHz bandwidth which exceeds B_c, thus an equalizer would be needed for this channel.

Small-Scale Fading
(Based on multipath time delay spread)

Flat Fading
1. BW of signal < BW of channel
2. Delay spread < Symbol period

Frequency Selective Fading
1. BW of signal > BW of channel
2. Delay spread > Symbol period

Small-Scale Fading
(Based on Doppler spread)

Fast Fading
1. High Doppler spread
2. Coherence time < Symbol period
3. Channel variations faster than base-band signal variations

Slow Fading
1. Low Doppler spread
2. Coherence time > Symbol period
3. Channel variations slower than baseband signal variations

Small Scale Fading Models

- Two categories:
- Non line-of-sight (NLOS); LOS does not exist

  Rayleigh fading model

- Line-of-sight (LOS) exists

  Rician fading model

- Models provide the probability distribution function of the magnitude of voltage envelope at an instance
- To determine the receiver power at any instance, we need to multiply the received power without fading and shadowing by the power envelope given by fading model ; then we need to add the shadowing effect

Rayleigh Fading Model

- Used to describe the statistical time varying nature of wireless signal with flat fading
- The received voltage envelope is provided by the Rayleigh distribution as follows

$$p(r) = \begin{cases} \dfrac{r}{\sigma^2}\exp\left(-\dfrac{r^2}{2\sigma^2}\right) & (0 \le r \le \infty) \\ 0 & (r < 0) \end{cases}$$

, σ is the rms value of the received voltage signal before *envelope detection*

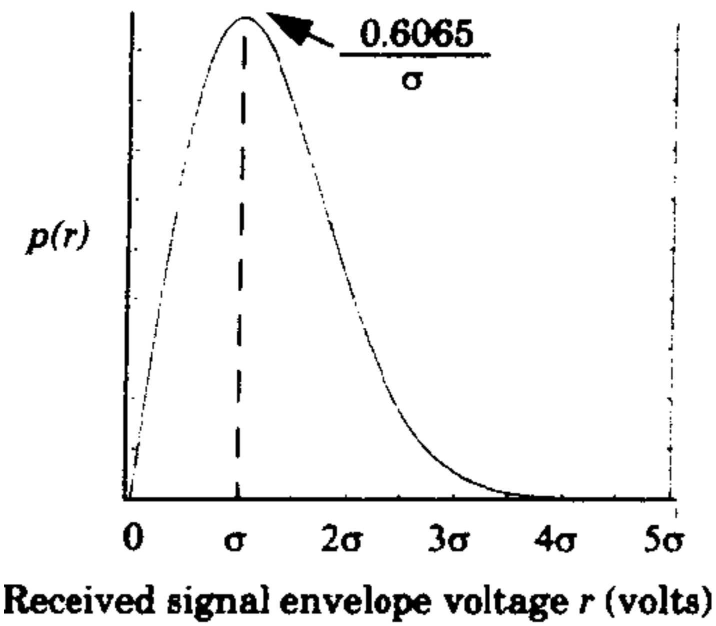
σ^2 **is the time-average power of the received signal**

which is the received power without fading and shadowing considering only log-distance path loss

Rayleigh Fading Model

$$Pr(r \le R) = \int_0^R p(r)\,dr = 1 - \exp\left(-\frac{R^2}{2\sigma^2}\right)$$

$$E[r] = \int_0^\infty r p(r)\,dr = \sigma\sqrt{\frac{\pi}{2}} = 1.2533\sigma$$



Ricean Fading Model

- When a stationary non-fading component, i.e., LOS component presents with the time varying nature of wireless signal with flat fading, then Ricean distribution is used
- The received voltage envelope is provided as follows

$$p(r) = \begin{cases} \dfrac{r}{\sigma^2} e^{-\frac{(r^2+A^2)}{2\sigma^2}} I_0\left(\dfrac{Ar}{\sigma^2}\right) & \text{for } (A \geq 0, r \geq 0) \\ 0 & \text{for } (r < 0) \end{cases}$$

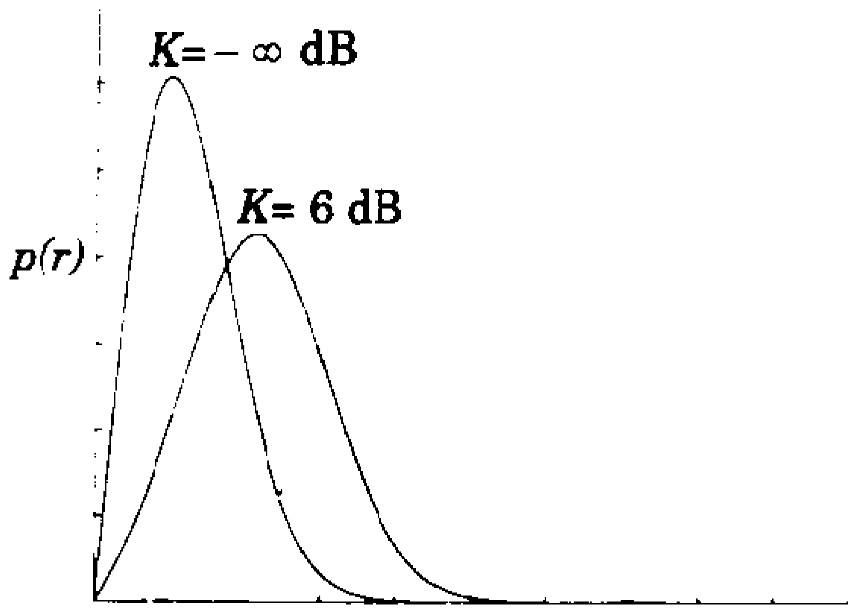
The parameter A denotes the peak amplitude of the dominant signal

$I_0(\cdot)$ is the modified Bessel function of the first kind and zero-order

Ricean Fading Model

ratio between the deterministic signal power and the variance of the multipath

$$K(\text{dB}) = 10\log\frac{A^2}{2\sigma^2}\ \text{dB}$$



Received signal envelope voltage r (volts)